Class Activity

Practice Task

Q-1 Analyze the following scenario and develop an object-oriented solution using Java file handling concepts.

The Examination Office of a university is in the process of modernizing its student grade reporting system. Currently, student grades are recorded manually, which is both time-consuming and prone to error. You have been asked to develop a basic object-oriented application in Java that automates this process. The system should generate a file that stores the final grades of students at the end of each semester. Each record in the file must include the student's name, roll number, subject, and grade. The program should overwrite the file each time it runs to ensure that only the latest report is saved. After writing the data, the program must read and display the report in a readable format. Your solution should apply object-oriented principles and include appropriate exception handling to deal with potential file-related errors.

Solution:

```
import java.io.File;

import java.io.FileWriter;

import java.io.IOException;

import java.util.Scanner;


// Student class to model student data

class Student {

    private String name;

    private String rollNumber;

    private String subject;

    private String grade;


    public Student(String name, String rollNumber, String subject, String grade) {

        this.name = name;

        this.rollNumber = rollNumber;

        this.subject = subject;
```

```
        this.grade = grade;
    }

    // Returns student data in a formatted string
    public String getFormattedRecord() {
        return "Name: " + name + ", Roll No: " + rollNumber +
                ", Subject: " + subject + ", Grade: " + grade;
    }
}

public class GradeReportGenerator {
    public static void main(String[] args) {
        try {
            // Define file path
            File reportFile = new File("GradeReport.txt");

            // Create the file (only first time)
            if (reportFile.createNewFile()) {
                System.out.println("File created: " + reportFile.getName());
            } else {
                System.out.println("File already exists. It will be overwritten.");
            }

            // Sample student data
            Student[] students = {
                new Student("Ayesha Khan", "23CS001", "Data Structures", "A"),
```

```java
            new Student("Usman Tariq", "23CS002", "Algorithms", "B+"),
            new Student("Fatima Ali", "23CS003", "Operating Systems", "A-")
        };

        // Write to the file (overwrite each time)
        FileWriter writer = new FileWriter(reportFile); // overwrite mode
        for (Student student : students) {
            writer.write(student.getFormattedRecord() + "\n");
        }
        writer.close();

        System.out.println("\nGrade Report Written Successfully.\n");

        // Read and display file content
        Scanner reader = new Scanner(reportFile);
        while (reader.hasNextLine()) {
            String line = reader.nextLine();
            System.out.println(line);
        }
        reader.close();
      } catch (IOException e) {
        System.out.println("An error occurred during file operations.");
        e.printStackTrace();
      }
    }
}
```